Configuração 8: $M = 128$, $C = 8$, $K = 12$, IoT = 20 dB, QPSK
Perda de SNR do BCD-MMSE (4 iterações) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.60 1.8%	0.95 2.1%	0.03 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.60 1.8%	0.69 2.4%	0.03 3.7%	0.00 4.2%
	AP_SAT_SYM	4.60 1.9%	0.95 2.2%	0.03 3.7%	0.00 4.2%
	AP_WRAP	4.60 1.7%	0.36 2.3%	0.03 3.5%	0.00 4.1%
	AP_WRAP_SM	4.60 1.8%	1.00 2.2%	0.03 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.93 2.3%	0.02 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.67 2.4%	0.02 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.93 2.5%	0.02 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.46 2.0%	0.02 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.99 2.4%	0.02 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.58 2.0%	0.94 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.58 1.8%	0.68 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.58 2.1%	0.94 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.58 1.7%	0.47 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	2.58 2.0%	1.00 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.58 2.0%	0.94 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.58 1.8%	0.68 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.58 2.2%	0.94 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.58 1.6%	0.47 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.58 2.0%	1.00 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.57 1.9%	0.94 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.57 1.8%	0.68 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.57 2.1%	0.94 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.57 1.8%	0.47 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.57 2.0%	1.00 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.57 2.0%	0.94 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.57 1.8%	0.68 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.57 2.2%	0.94 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.57 1.6%	0.47 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.57 2.0%	1.00 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.58 2.0%	0.94 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.58 1.8%	0.68 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.58 2.2%	0.94 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.58 1.8%	0.47 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.58 2.0%	1.00 2.4%	0.01 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.78 1.8%	0.92 2.1%	0.03 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.78 1.8%	0.72 2.4%	0.03 3.7%	0.00 4.2%
	AP_SAT_SYM	4.78 1.9%	0.92 2.2%	0.03 3.7%	0.00 4.2%
	AP_WRAP	4.78 1.7%	0.03 2.3%	0.03 3.5%	0.00 4.1%
	AP_WRAP_SM	4.78 1.8%	0.94 2.2%	0.03 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.92 2.3%	0.03 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.72 2.4%	0.03 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.92 2.5%	0.03 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.03 2.0%	0.03 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.93 2.4%	0.03 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.55 2.0%	0.92 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.55 1.8%	0.71 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.55 2.1%	0.92 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.55 1.7%	0.02 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	2.55 2.0%	0.93 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.56 2.0%	0.92 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.56 1.8%	0.71 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.56 2.2%	0.92 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.56 1.6%	0.02 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.56 2.0%	0.93 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.58 1.9%	0.92 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.58 1.8%	0.71 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.58 2.1%	0.92 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.58 1.8%	0.02 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.58 2.0%	0.93 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.56 2.0%	0.92 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.56 1.8%	0.71 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.56 2.2%	0.92 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.56 1.6%	0.02 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.56 2.0%	0.93 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.57 2.0%	0.92 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.57 1.8%	0.71 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.57 2.2%	0.92 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.57 1.8%	0.02 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.57 2.0%	0.93 2.4%	0.01 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	5.04 1.8%	1.59 2.1%	0.04 3.1%	0.00 3.6%
	AP_SAT_ZERO	5.04 1.8%	0.75 2.4%	0.04 3.7%	0.00 4.2%
	AP_SAT_SYM	5.04 1.9%	1.59 2.2%	0.04 3.7%	0.00 4.2%
	AP_WRAP	5.04 1.7%	0.33 2.3%	0.04 3.5%	0.00 4.1%
	AP_WRAP_SM	5.04 1.8%	1.01 2.2%	0.04 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	1.59 2.3%	0.04 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.74 2.4%	0.04 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	1.59 2.5%	0.04 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.33 2.0%	0.04 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	1.00 2.4%	0.04 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.75 2.0%	1.59 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.73 1.8%	0.74 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.75 2.1%	1.59 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.73 1.7%	0.32 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	2.73 2.0%	1.00 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.75 2.0%	1.59 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.73 1.8%	0.74 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.75 2.2%	1.59 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.73 1.6%	0.32 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.73 2.0%	1.00 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.79 1.9%	1.59 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.77 1.8%	0.74 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.79 2.1%	1.59 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.77 1.8%	0.32 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.77 2.0%	1.00 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.79 2.0%	1.59 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.77 1.8%	0.74 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.79 2.2%	1.59 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.77 1.6%	0.32 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.77 2.0%	1.00 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.75 2.0%	1.59 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.73 1.8%	0.74 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.75 2.2%	1.59 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.73 1.8%	0.32 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.73 2.0%	1.00 2.4%	0.01 3.7%	0.00 4.2%

TDLC300-100

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	4.91 1.8%	0.96 2.1%	0.04 3.1%	0.00 3.6%
	AP_SAT_ZERO	4.91 1.8%	0.64 2.4%	0.04 3.7%	0.00 4.2%
	AP_SAT_SYM	4.91 1.9%	0.96 2.2%	0.04 3.7%	0.00 4.2%
	AP_WRAP	4.91 1.7%	0.32 2.3%	0.04 3.5%	0.00 4.1%
	AP_WRAP_SM	4.91 1.8%	0.97 2.2%	0.04 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	× 2.0%	0.97 2.3%	0.10 3.7%	0.00 4.2%
	AP_SAT_ZERO	× 1.8%	0.72 2.4%	0.10 3.7%	0.00 4.2%
	AP_SAT_SYM	× 2.2%	0.97 2.5%	0.10 3.7%	0.00 4.2%
	AP_WRAP	× 1.6%	0.36 2.0%	0.10 3.2%	0.00 3.7%
	AP_WRAP_SM	× 2.0%	0.98 2.4%	0.10 3.7%	0.00 4.2%
AP_RND	AP_SAT	2.86 2.0%	0.95 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.86 1.8%	0.67 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.86 2.1%	0.95 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.86 1.7%	0.30 2.3%	0.01 3.5%	0.00 4.1%
	AP_WRAP_SM	2.86 2.0%	0.96 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	2.86 2.0%	0.95 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.86 1.8%	0.67 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.86 2.2%	0.95 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.86 1.6%	0.30 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.86 2.0%	0.96 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	2.86 1.9%	0.95 2.2%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.86 1.8%	0.67 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.86 2.1%	0.95 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.86 1.8%	0.30 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.86 2.0%	0.96 2.3%	0.01 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	2.86 2.0%	0.95 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.86 1.8%	0.67 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.86 2.2%	0.95 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.86 1.6%	0.30 2.0%	0.01 3.2%	0.00 3.7%
	AP_WRAP_SM	2.86 2.0%	0.96 2.4%	0.01 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	2.86 2.0%	0.95 2.3%	0.01 3.7%	0.00 4.2%
	AP_SAT_ZERO	2.86 1.8%	0.67 2.4%	0.01 3.7%	0.00 4.2%
	AP_SAT_SYM	2.86 2.2%	0.95 2.5%	0.01 3.7%	0.00 4.2%
	AP_WRAP	2.86 1.8%	0.30 2.4%	0.01 3.7%	0.00 4.2%
	AP_WRAP_SM	2.86 2.0%	0.96 2.4%	0.01 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	2.96 1.7%	2.55 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	2.94 1.6%	2.50 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	2.96 1.8%	2.55 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	2.87 1.6%	2.41 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	3.05 1.7%	2.68 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	× 1.9%	2.65 2.3%	0.05 3.9%	0.00 4.4%
	AP_SAT_ZERO	× 1.7%	2.61 2.2%	0.05 3.9%	0.00 4.4%
	AP_SAT_SYM	× 2.1%	2.65 2.5%	0.05 3.9%	0.00 4.3%
	AP_WRAP	× 1.4%	2.51 1.9%	0.05 3.4%	0.00 3.9%
	AP_WRAP_SM	× 1.9%	2.79 2.3%	0.05 3.9%	0.00 4.4%
AP_RND	AP_SAT	2.94 1.8%	2.55 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.90 1.7%	2.50 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.94 2.0%	2.55 2.4%	0.00 3.9%	0.00 4.4%
	AP_WRAP	2.84 1.6%	2.42 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	3.05 1.9%	2.69 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	2.94 1.9%	2.55 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.90 1.7%	2.50 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.94 2.1%	2.55 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.84 1.4%	2.42 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.05 1.9%	2.69 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	2.94 1.8%	2.55 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.90 1.7%	2.50 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	2.94 2.0%	2.55 2.4%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.84 1.6%	2.42 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	3.05 1.9%	2.69 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	2.94 1.9%	2.55 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.90 1.7%	2.50 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.94 2.1%	2.55 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.84 1.4%	2.42 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	3.05 1.9%	2.69 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	2.94 1.9%	2.55 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	2.90 1.7%	2.50 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	2.94 2.1%	2.55 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	2.84 1.6%	2.42 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	3.05 1.9%	2.69 2.2%	0.00 3.9%	0.00 4.4%